Alpha Prototype Progress Check

Cold Brew Case | Team: Bob the Builder

Team Members

Name	Roles	E-mail & GitHub ID
Khushi Mehta	*Team Captain, Research & game guidance*	*khushipr@usc.edu* *Khushim14*
Kavya Sharma	*Analytics, User Interface*	*kavyajay@usc.edu* *kavyaaaa16*
Ridham Patel	*Product Manager*	*ridhamas@usc.edu* *ridham2003710*
Karna Mehta	*Research and Development*	*karnamehta8@gmail.com* *Karna2701*
Aryan Shah	*Game Feel, Guidance*	*aryanutp@usc.edu* *aryan1410*

Playable build: https://rcodeborg2311.github.io/Cold_Brew_Case_Alpha/

Current repository: https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha

GitHub Classroom repository: [Add the assigned Classroom repository URL; confirm code is pushed there.]

Logline

Use directional footsteps and temporary coffee scans to track a moving thief, find a clear net shot around cover, and capture it before a short timer runs out.

Game Description

Cold Brew Case is a top-down 2D stealth pursuit game in Unity 6000.3.22f1 and C#. Courtyard, Corridor, Fork and Hideout form the current graybox. An opening chase leads into a 96-second hunt. Footstep pickups give short directions. Coffee grants up to 10 seconds of scanning: mouse aim selects one connected neighbor to reveal. Other rooms stay covered outside scanning. An 18-second shake follows, creating a tradeoff between gathering information and aiming accurately.

The detective avoids guards and positions around cover. In the same room, the thief becomes visible and flees or seeks cover after clear sight. Two nets stop at the first solid wall, block or guard. Misses cost a net and 9.6 seconds; pushing costs 14.4 seconds; spotting costs 19.2 seconds and returns the player to a checkpoint. Capture wins, including with the last net; a final miss or time expiry loses. No net throws during scanning or new coffee while shaking.

Mechanic Research

Member 1

Researcher / game / mechanic: Khushi Mehta / Criminal Case / Energy-limited, timed crime-scene investigation

Mechanic or system of interest: Each crime-scene search costs energy, and finding clues faster earns a higher score. The clues you collect lead step by step to arresting a suspect.

Why it interests our team: It pairs investigation with a limited resource and time pressure, much like our detective spending coffee to gather clues against the clock.

Interaction with other mechanics: Energy decides when you can search, speed affects your score, and hints help but lower it. Each clue narrows down the suspects.

Genre twist and innovation: We turn the static investigation into real-time stealth with a thief who moves. Coffee scans replace energy, and the shake afterward is the cost.

Play evidence and source:
https://apps.apple.com/us/app/criminal-case-hidden-objects/id767473889

Member 2

Researcher / game / mechanic: Ridham / *Mark of the Ninja* / Stealth tool selection.

Mechanic or system of interest: Players choose tools and techniques that support different approaches to stealth encounters

Why it interests our team: It helps me design Level 1 around meaningful tool choices and plan analytics to evaluate player decisions.

Interaction with other mechanics: Tools work with movement and positioning; their usefulness depends on how players approach opponents.

Genre twist and innovation: Our coffee scan provides temporary information but causes shaking afterward, making players balance visibility against net accuracy.

Play evidence and source: [https://store.steampowered.com/app/214560/Mark_of_the_Ninja]

Member 3

Researcher / game / mechanic: Kavya Sharma / Among Us / Information gathering and deduction

Mechanic or system of interest: Players gather information by observing the movements of other players, checking cameras, comparing what they observe with the events. Since the players have incomplete information, they need to interpret evidence

Why it interests our team: this demonstrates how limited information would encourage a player to observe surroundings and investigate. That's how our detective follows footsteps and does coffee scans to check rooms.

Interaction with other mechanics: information gathering interacts with movement, player actions and timing of other events. Our coffee scan reveals a selected neighboring room temporarily, while movement, guards, and the countdown continue to affect the detective's decisions.

Genre twist and innovation: Our player uses directional clues and time-bound scans to find the moving thief.

Play evidence and source: https://www.innersloth.com/games/among-us/

Member 4

Researcher / game / mechanic: Karna Mehta / *Penguin Diner* / Time management and customer patience

Mechanic or system of interest: Players seat customers, take orders, serve food, and clean tables while managing a limited shift time. Customers lose patience while waiting, so players must prioritize tasks to keep service moving.

Why it interests our team: It shows how a countdown and competing tasks make the player choose what to do next. Our detective faces a similar choice between following clues, scanning a room, and trying to catch the thief before time runs out.

Interaction with other mechanics: Task priority depends on customer patience, ready orders, and open tables. In our game, the timer, guard pressure, footsteps, coffee scans, and nets all affect the detective's next move.

Genre twist and innovation: We adapt time-management pressure into a real-time detective chase. Instead of serving customers before they leave, the detective must use clues and tools efficiently to catch a moving thief.

Play evidence and source: Play *Penguin Diner* on Coolmath Games

Member 5

Researcher / game / mechanic: Aryan / Alien: Isolation / Motion tracker
Mechanic or system of interest: Players carry a handheld tracker that shows nearby moving threats with pings that speed up as danger gets closer. It only shows movement, so players must interpret vague signals. Raising it also limits what else the player can do.
Why it interests our team: It shows information that has a cost. The tracker helps players decide where to go, but using it leaves them exposed. That matches our coffee scan, which reveals a room briefly and then causes shaking. Its escalating audio cues also inform my guidance work.
Interaction with other mechanics: The tracker interacts with movement, noise and enemy behavior. Players slow down, check it, then cross open space. In our game, scanning blocks net throws, the shake lowers aim, and guards and the countdown still pressure the detective.
Genre twist and innovation: We turn "where is the threat?" into "which room is the thief in?" through short, player-selected scans of one neighboring room. The delayed shake makes each scan a commitment.
Play evidence and source: https://store.steampowered.com/app/214490/Alien_Isolation/

Mechanics Matrices

Twist and Mechanics Matrix

The core twist is temporary information followed by reduced aim stability. These entries describe the current four-room checkpoint.

Mechanic	Implemented behavior	Connection to twist	Genre element	Innovation
Footsteps	One-shot direction and two thumps; old hint clears on room change.	Helps choose where to explore or scan.	Tracking	Short route guidance
Coffee scan	F spends one coffee for up to 10 seconds; aim chooses one neighbor.	Cross-room visibility exists only during the effect.	Information; resources	Player-selected timed visibility
Shaking	18 seconds of decaying aim wobble after scan.	Nets need scan to end; new coffee waits until shake ends.	Accuracy; timing	Observation has a later cost
Guards and push	Patrols/cones; catch returns player; push makes noise.	Danger and time costs shape movement between scans.	Stealth; avoidance	Information competes with urgency
Thief and cover	Clear sight triggers fleeing or cover seeking in the same room.	Locating the thief still requires a clear capture angle.	Pursuit; positioning	Reactive final encounter
Nets	Two shots; nearest solid hit blocks the throw.	No shots during scan; shaky shots remain possible.	Tools; line of sight	Limited attempts make positioning matter
Time left	96-second hunt plus push, catch and miss penalties.	Travel, scan and shake waiting consume the budget.	Urgency	Shared pressure across decisions

Controls: WASD move; mouse aim; left click net; F scan/end; Space push; P action pause.

Mechanic Interaction Matrix

Affects →	Footsteps	Scan and shake	Guards and push	Thief, cover and nets	Time left
Footsteps	—	Guides where to look	Guides movement routes	Guides pursuit	Following hints uses time
Scan and shake	No cue required to scan	—	Can expose thermal guards	Scan blocks shots; shake wobbles aim	Scan and waiting use time
Guards and push	—	Being caught ends active scan	—	Guards block nets	Catch / push apply penalties
Thief, cover and nets	Cue targets next route step	Throw needs scan off	A guard hit prevents capture	—	Miss costs time; capture ends hunt
Time left	Limits detours	Limits scan / wait decisions	Penalties reduce budget	Expiry prevents capture and triggers escape	—

Initial Mechanics Diagram

The checkpoint flow shows tracking, optional scanning, shaking, cover and limited net attempts. It illustrates the implemented four-room game.



Weekly Prototype Descriptions

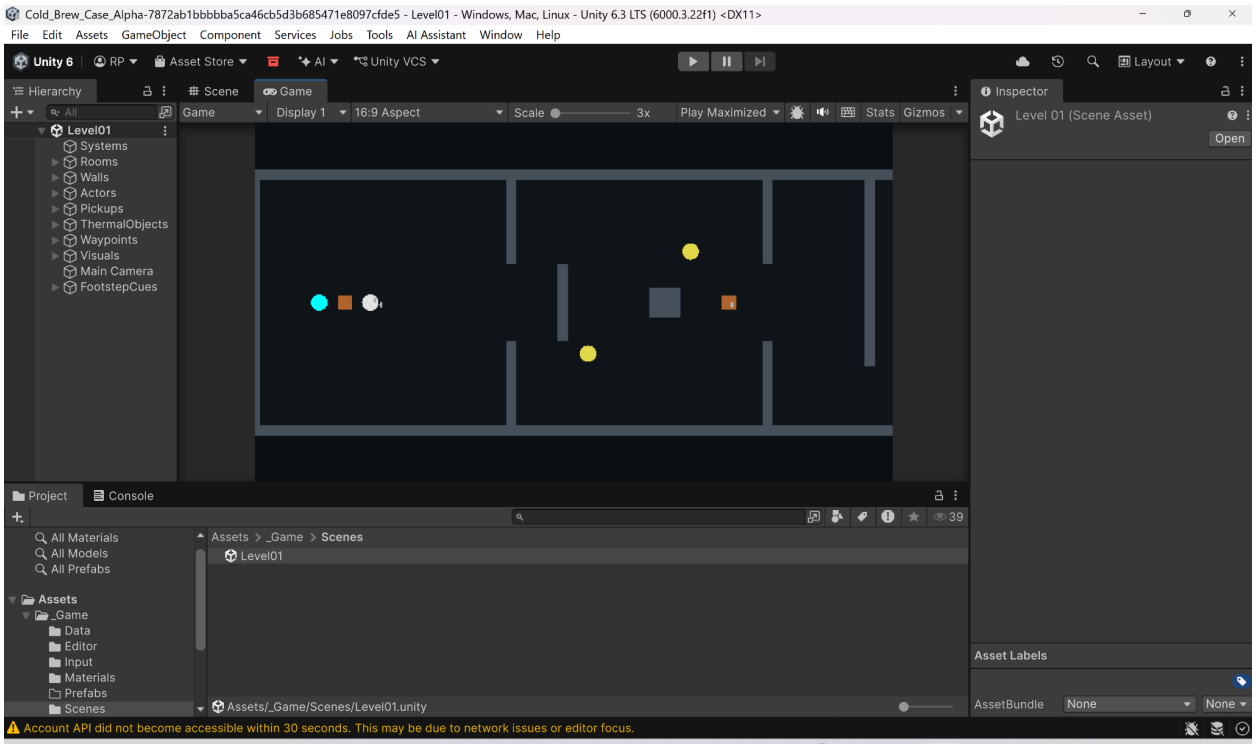
Week 1

Dates and build or commit: [OCT 05 to OCT 09] / https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/7872ab1bbbbba5ca46cb5d3b685471e8097cfde5

|

Prototype and changes: The level was playable in a basic game setting sense but game design wasn't complete.

Screenshot:



What the image demonstrates: Shows a raw first version of the game

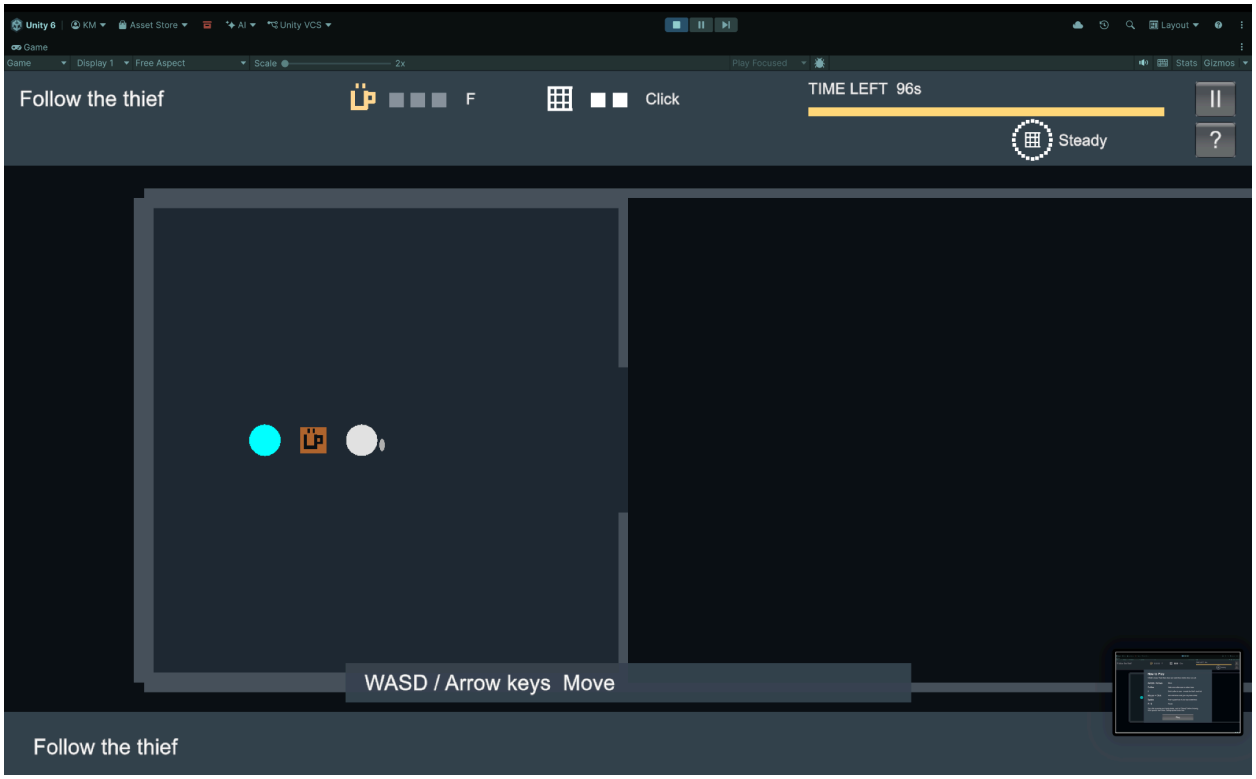
Feedback and next change: To reduce confusion and reduce game learning curve.

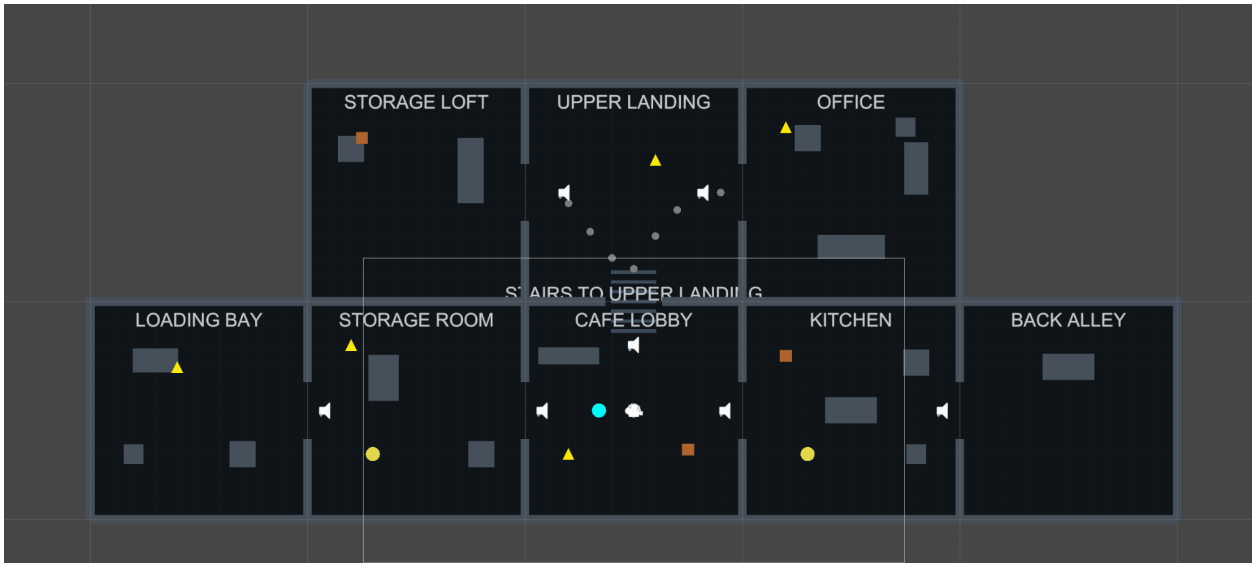
Week 2

Dates and build or commit: [OCT 09 to OCT 13] / https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/c74138aff871532d9717255a63ca5e7191ae0976

Prototype and changes: Level 1 is playable now, and it teaches the mechanics of the game

Screenshot:


Follow the thief
F
Click
TIME LEFT 96s
Steady
WASD / Arrow keys Move
Follow the thief


STORAGE LOFT
UPPER LANDING
OFFICE
LOADING BAY
STORAGE ROOM
CAFE LOBBY
KITCHEN
BACK ALLEY

Weekly Progress

Week 1 Individual Contributions

Member 1	Member 2	Member 3	Member 4	Member 5
[Name]	Ridham	Kavya	Karna	Aryan
Worked on Level 1, fixed player movement and added in-game instructions.	Build First Level version 1	Fixed issues with throwing net	Built and designed the level 2 player and thief mechanics. Improved the room designs and tested.	Level 2 hints and guidance
https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/cde70a272d4495bc97049f886aff18efd4b779d1	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/7872ab1bbbbba5ca46cb5d3b685471e8097cfde5	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/811c77cb8afa986896493d3f0dcf056bb21a0f21	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/pull/2/changes/5d020efba37fb495ea24f4b9d177bcb7a71f6538	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/dc0d47e5be2d1bad6b0d4a9f67f97e1027b1a70b
Tested that movement was smooth and responsive and that instructions appeared clearly in Level 1.	Tested Player Mechanics and Graybox	Tested the fix works	Tested the Level 2 design.	Hints appearing at defined time intervals of inactivity

Week 2 Individual Contributions

Member 1	Member 2	Member 3	Member 4	Member 5
[Name]	Ridham	Kavya	Karna	Aryan
Built the thief mechanics: movement, fleeing Also wrote the GitHub documentation (README and screenshots).	Completed level 1 and published it.	Improved Level 2 thief speed, guard patrols, and footstep tracking.	Used level 1 review to plan how Level 2 could extend the existing project while preserving Level 1 behavior.	Added low-time guidance tip to Level 2.

Member 1	Member 2	Member 3	Member 4	Member 5
https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/cde70a272d4495bc97049f886aff18efd4b779d1	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/c74138aff871532d9717255a63ca5e7191ae0976	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/pull/3/changes/a3ba377f93395273692f3e476aa02f7e6457f040 https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/pull/3/changes/a6efa5fbb49d5b2ae24aba753b315ba4ce6e61d5	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/1eba0fd689a4c4d1767c5280406362c6620fd65e	https://github.com/rcodeborg2311/Cold_Brew_Case_Alpha/commit/9ef4023c5ebe8a49e828b0b4c877707cd3d30084
Tested the changes	Tested latency and publish settings	Tested new changes	Tested the existing changes and end-to-end UX flow.	Tested low-time guidance tip

External References

These resources supported the code, setup or documentation. Add each member's research-game sources and any other resource actually used.

Unity 6.3 manual — Engine and component reference for Unity 6000.3.22f1.

Mark of the Ninja — Official description consulted for movement/tool stealth reference.

Invisible Inc — Official description consulted for guarded-space decision reference.

Heat Signature — Official description consulted for gadget and improvisation reference.

Among Us — Official Website — Reference for information gathering, observation, and deduction mechanics in a social deduction game.

Alien: Isolation - Reference for motion tracker mechanic

Criminal Case — Official App Store listing, used as a reference for energy-limited, timed crime-scene investigation.

Play *Penguin Diner* on Coolmath Games - Official Gaming Guide and Installation.